AI Career Companion

Aditya Ashok
University of Illinois,
Urbana-Champaign
aashok8
aashok8@illinois.edu

Simran Malhotra
University of Illinois,
Urbana-Champaign
simranm2
simranm2@illinois.edu

Sitara Achar
University of Illinois,
Urbana-Champaign
sitaraa2
sitaraa2@illinois.edu

Yug Vajani
University of Illinois,
Urbana-Champaign
yvajani2
yvajani2@illinois.edu

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Abstract
This report introduces AI Career Companion, an interactive system designed to address the fundamental disconnection between job seekers and effective resume feedback mechanisms. While numerous resume optimization tools exist in the market, most focus primarily on keyword matching rather than providing personalized, contextually relevant guidance. Our work leverages advances in large language models combined with human-centered design principles to create a more interactive, transparent, and iterative resume feedback experience. Through careful prompt engineering, rule-based expert knowledge integration, and an interactive user interface, AI Career Companion delivers tailored resume analysis, skill gap identification, and career guidance that aligns with human judgment and professional hiring standards. User evaluations suggest that this approach significantly enhances the resume development process by enabling dialogue-based refinement rather than one-way analysis. This project contributes to the growing body of research on human-AI collaboration systems that prioritize user agency and meaningful interaction.

Introduction
The process of crafting an effective resume remains one of the most challenging aspects of professional development. Despite technological advances in recruitment systems, job seekers continue to struggle with understanding how to

present their qualifications effectively for specific opportunities. This challenge is particularly acute given that resume expectations vary significantly across industries, roles, and organizational contexts. Current technological solutions often fail to bridge this knowledge gap, leaving job seekers to navigate complex application processes with insufficient guidance.

Existing resume tools typically optimize for keyword matching to pass Applicant Tracking Systems (ATS) but rarely provide the contextual, personalized feedback that genuinely advances a candidate's application. Most critically, these systems lack the interactive dimension that characterizes effective career coaching relationships—the ability to engage in an iterative dialogue about strengths, weaknesses, and strategic presentation choices.

AI Career Companion addresses this limitation by re-imagining the resume feedback process through the lens of human-AI collaboration. We approach resume development not as a mechanical optimization problem but as a nuanced communication challenge that benefits from conversational exploration. By combining advanced language model capabilities with carefully structured expert knowledge and interactive design elements [2] [3], our system creates an experience that more closely resembles consulting with a knowledgeable career advisor than submitting to an automated checker. Crafting effective prompts can be challenging and prompt-based interactions are brittle [8]. Despite their potential, the lack of need for structured input can make it more challenging for users to translate their intentions into actions accurately. [10] To tackle this, AI Career Companion integrates carefully engineered based on thorough experimentation and research to aid users in maximizing their efficacy while interacting with AI for this specific task at hand.

This report details our approach to developing AI Career Companion, including its theoretical foundations, technical implementation, and initial user feedback. We examine how principles from human-computer interaction research can inform more effective AI-assisted career development tools, with particular emphasis on transparency, personalization, and user agency in the feedback process.

Related Work

The development of AI Career Companion builds upon several important strands of prior research in human-computer interaction, machine learning, and career development systems. Understanding these connections helps situate our work within the broader research landscape.

Current commercial resume development tools such as Jobscan and ResumeWorded offer valuable services for ATS optimization and targeted resume building. However, as noted by Wang et al. in their work on conversational interaction with mobile UI using Large Language Models, these systems typically lack the natural conversational flow that characterizes effective human coaching relationships. Our approach draws inspiration from their findings on prompt structure optimization for generative AI systems, particularly regarding how to elicit coherent, contextually appropriate responses [1].

The theoretical framework for our interactive approach is significantly influenced by Amershi et al.'s "Power to the People: The Role of Humans in Interactive Machine Learning," which emphasizes the importance of human agency and iterative feedback loops in AI system design. Their work on co-adaptive learning systems provides a valuable foundation for understanding how resume feedback tools can evolve through user interaction [3].

Brown et al.'s seminal work on few-shot learning capabilities in language models provides technical grounding for our approach to prompt engineering and contextual adaptation. Their demonstration that properly structured prompts can guide language model performance without explicit retraining has directly influenced our strategy for injecting expert resume knowledge into AI-generated feedback [2].

Our methodology also draws from Martelaro and Ju's "The Needfinding Machine," which presents techniques for using conversational agents as probes to understand user needs and reactions. By implementing a similar approach, AI Career Companion functions not only as a feedback tool but also as a mechanism for ongoing discovery about user requirements in the resume development process [4].

Vaccaro et al.'s "Designing the Future of Personal Fashion" offers particularly relevant insights, despite its focus on a different domain. Their critique of "black-box" recommendation systems and emphasis on multi-faceted personalization closely parallels our concerns about traditional resume tools. Both their work and ours emphasize the importance of user-involved design cycles to increase interpretability and trust—essential factors in career guidance systems [5].

Zamfirescu-Pereira et al.'s *"Why Johnny Can't Prompt: How Non-AI Experts Try (and Fail) to Design LLM Prompts"* provides critical insights into the challenges non-expert users face when interacting with large language models. Their study reveals that individuals without AI expertise often struggle with constructing effective prompts, leading to suboptimal interactions with LLMs. [8] Similarly, Rashon Poole's "Improving Human-LLM Interactions by Redesigning the Chatbot" offers valuable insight on how a lot of the current Natural Language Interfaces (NLI) systems do not incorporate key human-LLM design practices that help bridge the gap between a user's intention and actions. [10] By examining failures in generic chatbot interactions and proposing redesigns rooted in user-centered design principles, the work highlights the importance of clear intent modeling and scaffolding user prompts to minimize ambiguity in AI responses. These principles align closely with the AI Career Companion's interface strategy, particularly our focus on goal-driven dialogue structures and adaptive prompt suggestions. These findings underscore the necessity for AI systems to incorporate user-friendly interfaces and guidance mechanisms. In developing AI Career Companion, we have integrated structured prompts and contextual assistance to mitigate these challenges, aiming to enhance user engagement and effectiveness in career planning tasks. Poole's emphasis on transparency and iterative refinement also directly supports our goal of creating a conversational agent that not only provides actionable resume feedback but also improves with continued user interaction.

Zinjad et al.'s 'ResumeFlow: An LLM-facilitated Pipeline for Personalized Resume Generation and Refinement' presents a compelling demonstration of how large language models (LLMs) can be operationalized to facilitate personalized resume generation. Their pipeline—a modular system comprising user data extraction, job description parsing, and targeted resume refinement—achieved strong user satisfaction scores across multiple human-subject evaluations, particularly in relevance, conciseness, and personalization. We draw inspiration from ResumeFlow's success in building a scalable, zero-shot LLM-based approach that uses prompt engineering to guide resume tailoring without requiring model fine-tuning. Like ResumeFlow, AI Career Companion seeks to improve the resume customization process through user-centric design and the strategic deployment of LLM capabilities. However, our system expands on this foundation by integrating iterative user feedback, implementing prompt scaffolding informed by usability research, and offering fine-tuned LLM capabilities for more domain-specific resume diagnostics. This hybrid approach enhances the system's adaptability to diverse user contexts and provides a richer interactive experience aligned with career development goals. [9]

Finally, our implementation strategy is informed by industry guidelines such as Google PAIR's "People + AI Guidebook" [6] and Microsoft's "Guidelines for Human-AI Interaction" [7]. These frameworks provide practical direction for aligning feedback mechanisms with model capabilities and building user trust through transparent feedback loops. Particularly influential were the Microsoft guidelines surrounding "learning from user behavior and updating cautiously," which informed our implementation of user feedback surveys and follow-up prompt functionality.

Research Question

The central research question guiding this project is: How can interactive AI systems deliver transparent and personalized resume feedback that aligns with human judgment and hiring standards?

This question addresses a significant gap in current career development technologies, which typically prioritize algorithmic optimization over human-centered guidance. Our investigation focuses specifically on the intersection of advanced language models and interactive design principles, exploring how their combination can create more effective career support systems.

Research Contributions

The primary contributions of this work include:

- A novel framework for interactive resume feedback that transcends the limitations of traditional keyword-matching systems, enabling dynamic, dialogue-based refinement of application materials.
- A hybrid intelligence approach that deliberately combines the pattern recognition capabilities of large language models with structured expert knowledge, ensuring feedback that is both creative and grounded in professional standards.
- An implementation strategy for embedding transparency into AI-generated career guidance, providing users with clear rationales for recommendations rather than opaque directives.
- A user-centered methodology for continuous system refinement through feedback collection and prompt engineering iteration, demonstrating how AI career tools can evolve through collaborative usage.
- Empirical insights from initial user testing regarding the comparative advantages of conversational resume feedback over traditional static analysis approaches.

By addressing these dimensions, our work contributes to the broader understanding of how AI systems can more effectively support complex professional development tasks that traditionally relied on human expertise. Rather than attempting to replace human career coaches, AI Career Companion demonstrates how technology can create accessible, scalable support systems that preserve the interactive qualities that make human guidance valuable.

AI Career Companion

AI Career Companion represents a paradigm shift in how job seekers approach resume development and job application preparation. At its core, this tool addresses the fundamental gap between static resume feedback systems and

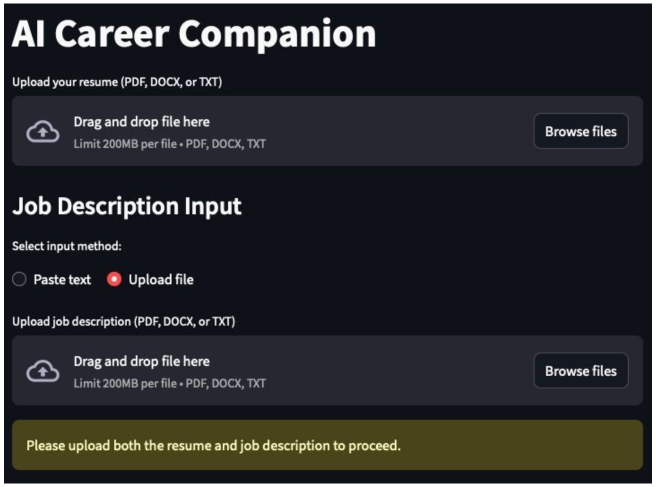

Figure 1: AI Career Companion UI

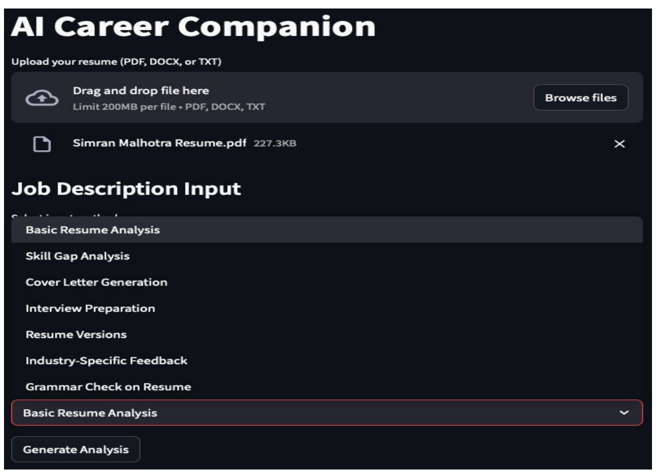

Figure 2: Analysis Types

the dynamic, personalized guidance that job seekers actually need.

Unlike conventional resume optimization tools that primarily focus on keyword matching for Applicant Tracking Systems (ATS), AI Career Companion creates a genuinely interactive experience that evolves with the user's needs. The system functions as a collaborative partner rather than a one-dimensional feedback mechanism. The platform enables users to upload their resumes and job descriptions in multiple formats (PDF, DOCX, TXT) and then select from a range of specialized feedback options. These include resume-job matching assessment, skill gap analysis, cover letter generation assistance, interview preparation guidance, industry-specific feedback, and even grammar and writing style evaluation.

What truly distinguishes AI Career Companion is its conversational capability. After receiving initial feedback, users can engage in follow-up dialogue, asking questions, seeking clarification, or requesting additional suggestions—creating an experience that mirrors consulting with a human career coach rather than a rigid automated system.

The tool deliberately prioritizes transparency, providing not just suggestions but also clear rationales for each recommendation. This approach helps users understand the "why" behind career advice, thereby building trust and facilitating deeper learning about effective application strategies. By bridging advanced AI capabilities with human-centered design principles, AI Career Companion transforms resume building from a frustrating technical exercise into an enlightening professional development opportunity.

Methodology

Our development methodology combined rigorous prompt engineering with human-centered design principles to cre-

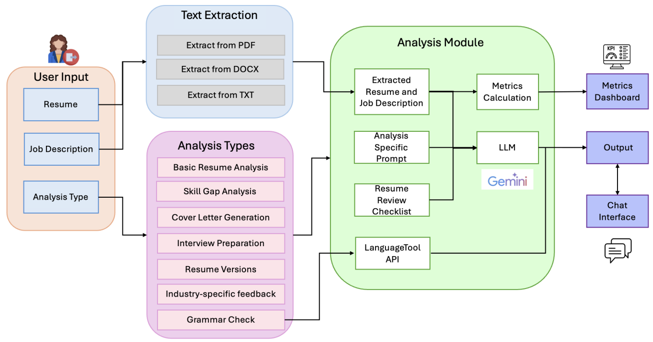

Figure 3: System Architecture

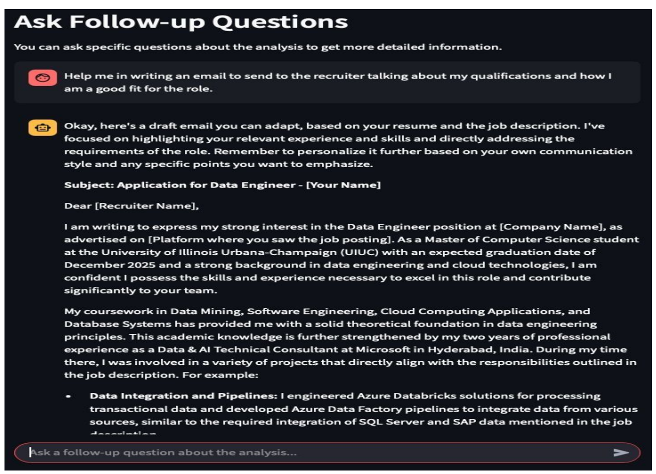

Figure 4: Chat Interface

ate a system that delivers consistently valuable feedback while maintaining the flexibility to adapt to diverse user needs.

Interactive Architecture
Rather than implementing a traditional one-way analysis model, we deliberately designed an architecture that places user exploration at the center. The Streamlit-based frontend provides multiple pathways for engagement, allowing users to self-direct their experience based on their specific career development priorities.

Hybrid Intelligence Approach
The system's intelligence framework deliberately combines two complementary approaches:

- LLM-Powered Analysis: We selected Gemini 2.0 Flash as our core language model due to its superior performance in generating structured, contextually aware feedback. The model excels at understanding nuanced relationships between resume content and job requirements, enabling sophisticated pattern recognition beyond what rule-based systems alone could achieve.
- Rule-Based Expert Knowledge: To ground the system in established best practices, we embedded a comprehensive Resume Review Checklist directly into our prompts. This deliberately constrains the LLM's outputs to align with expert validated resume standards while maintaining natural language fluidity.

Prompt Engineering Strategy
Our prompt design underwent multiple iterations to optimize for both performance and user experience:

- Role-Based Framing: Each prompt begins with explicit role definition (e.g., "You are an expert career coach") to establish appropriate tone and expertise level.
- Structured Output Requirements: Prompts include explicit instructions for formatting feedback into clear, digestible sections.
- Dynamic Context Injection: The system seamlessly integrates the user's resume and job description into each prompt, ensuring all recommendations remain contextually relevant.
- Two-Phase Analysis: For industry-specific feedback, we implemented a sequential prompting approach—first identifying the relevant industry, then tailoring subsequent guidance to that specific sector's conventions.

Transparency Through Explanation
Every recommendation generated by AI Career Companion includes accompanying rationales. This design decision serves multiple purposes:

- It helps users understand the reasoning behind suggestions rather than presenting them as arbitrary directives. grounded in recognizable professional principles.
- It supports learning by transforming feedback from simple corrections into educational opportunities.

Grammar Check Implementation
the LanguageTool API as a specialized component for identifying grammatical and stylistic issues that might otherwise escape detection in broader analysis.

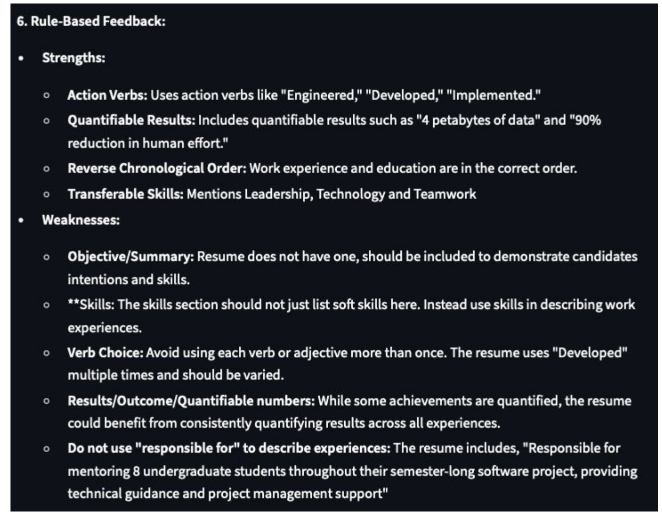

Figure 5: Rule-based Feedback

This methodology reflects our commitment to developing a system that balances technological sophistication with genuine utility for users navigating the complex process of career advancement.

Evaluation

To assess the performance and effectiveness of AI Career Companion, we conducted a multi-pronged evaluation covering both quantitative and user-centered dimensions. Our evaluation strategy includes (1) metrics-based resume quality assessment, (2) semantic relevance scoring using BERT Score across multiple models, and (3) a structured user study that informs system refinement through real-world feedback.

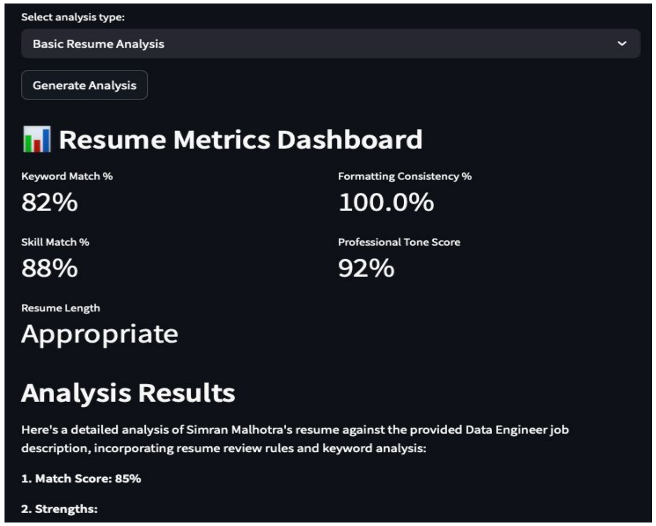

Figure 6: Resume Metrics Dashboard

Resume Metrics Dashboard
To provide users with tangible insights into their resume quality, we implemented a dashboard that evaluates five critical aspects:

- Keyword Match (%): Assesses overlap between resume and job description terminology using LLM-assisted semantic similarity.
- Skill Match (%): Estimates alignment with role-specific skills, factoring in both explicit mentions and implied capabilities.
- Formatting Consistency (%): Analyzes indentation and layout uniformity to ensure a professional appearance.

BERT Scores				
Role	Analysis Types	AI Career Companion Tool	Gemini 2.0 Flash	GPT 4o
Data Engineer Intern	Basic Resume Analysis	**80.99**	78.41	77.88
	Skill Gap Analysis	**80.26**	77.66	78.23
	Cover Letter Generation	**83.83**	79.05	79.71
	Resume Versions	**86.95**	77.31	77.52
Digital Marketing Manager	Basic Resume Analysis	81.45	80.82	**82.02**
	Skill Gap Analysis	**82.45**	80.37	80.99
	Cover Letter Generation	**83.92**	83.58	83.21
	Resume Versions	86.34	85.46	**86.53**
Software Engineer	Basic Resume Analysis	**81.07**	79.55	79.74
	Skill Gap Analysis	**80.22**	78.25	78.94
	Cover Letter Generation	80.67	79.99	**82.04**
	Resume Versions	**90.3**	85.4	86.7
Product Manager Intern	Basic Resume Analysis	76.4	75.78	**76.99**
	Skill Gap Analysis	**75.98**	74.28	75.65
	Cover Letter Generation	76.64	74.25	**76.78**
	Resume Versions	**80.92**	79.86	80.13

Figure 7: BERT Scores for Semantic Evaluation

- Professional Tone Score: Generated via LLM prompt to evaluate tone appropriateness on a scale of 0–100.
- Resume Length Classification: Flags resumes that are too short (<250 words) or too long (>900 words), based on industry norms.

Semantic Evaluation Using BERT Score
We used BERT Score, a state-of-the-art semantic similarity metric, to compare the system's outputs with the original job descriptions and resumes uploaded across multiple analysis types and roles. This helped evaluate how closely the generated content aligns with role expectations.

We compared responses from three systems:

- AI Career Companion (our tool)
- Gemini 2.0 Flash (baseline LLM)
- ChatGPT 4o

Across all roles and analysis types, AI Career Companion consistently outperformed the baseline Gemini 2.0 Flash model in terms of semantic similarity with job descriptions and resumes. These improvements stem from prompt customization, embedded resumewriting rules, and more job-specific response structuring.

While AI Career Companion excels at targeted, rule-based resume feedback, ChatGPT 4.0's broader language capabilities and training might give it an advantage in tasks requiring creative content generation and stylistic flexibility. As a result, ChatGPT 4o did outperform our system in a few cases (e.g., Resume Versions for Digital Marketing Manager). However, the overall results show stronger performance from our tool. This indicates that AI Career Companion tool offers higher fidelity in tailoring its responses to specific job expectations.

As part of future work, we plan to explore integrating GPT models into our backend to further improve accuracy and maintain competitive performance in scenarios where ChatGPT currently excels.

User Study

To assess real-world impact and usability, we conducted a user study with 45 participants, guided by HCI principles of needfinding and co-adaptive system design. The feedback form captured both quantitative and qualitative input:

- Perceived Usefulness: Average rating of 8.7/10 for helping improve resume quality.
- Comparison with Other Tools: Rated 8.4/10 in outperforming tools like ChatGPT for job-specific feedback.
- Success Prompts: Users contributed successful prompt templates that were integrated into future prompt iterations.
- Open-Ended Feedback: Participants praised the structured, conversational tone,, and actionable insights; some suggested expanding ATS-specific tips.

In addition to numeric ratings, the user study uncovered valuable design insights, including

> "It'll be interesting to get recommended to similar roles or openings, and also resume tips for such roles."

> "Information on Resume Analysis based on Experience Level / Career Stage would be a good addition to this web application."

> "Great analysis and tool — pinpointed gaps in resume with respect to the JD and gave solid recommendations on how to tackle them."

> "Incremental prompt did have context of previous prompt; rewriting prompt in slightly different ways gave different results."

> "Maybe allow users to paste job descriptions directly instead of requiring file uploads."

User feedback was not treated as static validation, but as dynamic design data, in line with participatory design and HCI feedback loop models. These insights informed prompt refinement, UI simplification, and the addition of new analysis types such as industryspecific feedback.

Together, these three dimensions offer a comprehensive picture of system effectiveness, model fidelity, and user-centered value, positioning the AI Career Companion as a robust, iterative, and intelligent career support system.

Future Work

This section outlines both short-term enhancements that can be pursued within the scope of this project, as well as long-term opportunities to extend the system beyond its current implementation.

Within the Scope of This Project

To continue improving the quality and interactivity of the AI Career Companion application, the following enhancements are proposed:

- Prompt Refinement via User Contributions: Continuously update and improve prompt templates using high-quality prompts shared by users to generate more accurate and relevant outputs.
- Support for Multiple LLMs: Extend the system to support and compare various large language models (LLMs), such as LLaMA, GPT-4, and others, using evaluation metrics like BERT Score.
- UI and Feature Enhancements: Add frontend features such as editable resume previews and drag-and-drop upload functionality to improve the user experience.
- Feature and Prototype Iteration: Refine core analysis modules (e.g., skill gap analysis, cover letter generation) based on user feedback and support configuration options such as tone or seniority level.

Beyond the Scope of This Project

The following areas offer potential extensions for future development:

- Job and Resume Databases: Build anonymized datasets of job descriptions and resumes across industries to support more domain-specific analysis and model tuning.
- Feedback-Driven Refinement: Collect user feedback on successful job application outcomes and use this data to improve prompt and model performance.
- Retrieval-Augmented Generation (RAG): Explore RAG-based architectures to improve the contextual grounding of LLM responses using real-world reference material.

Suggested Additional Metrics

To enhance the Resume Metrics Dashboard, future iterations can include the following metrics:

- ATS Compatibility Score: Evaluates the resume's compliance with standard ATS parsing formats.
- Readability Score: Measures the reading level using tools such as the Flesch-Kincaid index.
- Experience Relevancy Score: Assesses alignment between past roles and current job responsibilities.

Conclusion

AI Career Companion redefines the landscape of resume feedback by combining the strengths of large language models with human centered design to deliver transparent, personalized, and interactive career support. Our system bridges the gap between static keyword matching tools and the nuanced guidance of a career coach, offering users not only actionable insights but also the rationale behind each recommendation.

Through rigorous prompt engineering, a hybrid intelligence architecture, and a commitment to iterative refinement via user feedback, we have demonstrated that AI systems can meaningfully support job seekers in their professional development journey. Empirical evaluation, including semantic scoring and user studies, reinforces the system's effectiveness and highlights its value over existing solutions.

Looking forward, AI Career Companion sets the foundation for broader, domain-specific, and context-aware career tools that evolve through human-AI collaboration. By embracing transparency, adaptability, and user agency, this system moves closer to fulfilling the vision of scalable, intelligent career coaching for all.

REFERENCES

[1] Bryan Wang, Vinitha Gummadi, Scott Carter, Jeffrey Nichols, and Ranjitha Kumar. Enabling conversational interaction with mobile UI using large language models. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*, Association for Computing Machinery, 2023. `https://doi.org/10.1145/3544548.3580688`

[2] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, et al. Language models are few-shot learners. In *Advances in Neural Information Processing Systems (NeurIPS)*, 33:1877–1901, 2020. `https://proceedings.neurips.cc/paper/2020/hash/1457c0d6bfcb4967418bfb8ac142f64a-Abstract.html`

[3] Saleema Amershi, Maya Cakmak, William Bradley Knox, and Todd Kulesza. Power to the people: The role of humans in interactive machine learning. *AI Magazine*, 35(4):105–120, 2014. `https://doi.org/10.1609/aimag.v35i4.2513`

[4] Nikolas Martelaro and Wendy Ju. The Needfinding Machine. In *Proceedings of the 2019 ACM Conference on Designing Interactive Systems (DIS '19)*, pages 1015–1027. ACM, 2019. `https://doi.org/10.1145/3322276.3322365`

[5] Kristen Vaccaro, Yulia Tsvetkov, Jessica Hullman, and Ranjitha Kumar. Designing the Future of Personal Fashion. In *Proceedings of the 31st Annual ACM Symposium on User Interface Software and Technology (UIST '18)*, pages 123–135. ACM, 2018. `https://doi.org/10.1145/3242587.3242662`

[6] Google PAIR. People + AI Guidebook: Designing with the user in mind. `https://pair.withgoogle.com/guidebook`

[7] Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, Paul Bennett, Kori Inkpen, Jaime Teevan, Ruth Kikin-Gil, Eric Horvitz Guidelines for Human-AI Interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*, Paper 3, 13 pages. ACM, 2019. `https://doi.org/10.1145/3290605.3300233` Microsoft Research. `https://www.microsoft.com/en-us/research/publication/guidelines-for-human-ai-interaction`

[8] J.D. Zamfirescu-Pereira, Richmond Y. Wong, Bjoern Hartmann, and Qian Yang. Why Johnny Can't Prompt: How Non-AI Experts Try (and Fail) to Design LLM Prompts. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems (CHI '23)*,

Article 437, pages 1–21. ACM, 2023. https://doi.org/10.1145/3544548.3581388

[9] Saurabh Bhausaheb Zinjad, Amrita Bhattacharjee, Amey Bhilegaonkar, and Huan Liu. ResumeFlow: An LLM-facilitated Pipeline for Personalized Resume Generation and Refinement. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR '24)*, pages 1–4. ACM, 2024. https://doi.org/10.1145/3626772.3657680

[10] Rashon Poole. Improving Human-LLM Interactions by Redesigning the Chatbot. *CS224N Final Project Report, Stanford University*, 2023. https://web.stanford.edu/class/archive/cs/cs224n/cs224n.1244/final-projects/RashonPoole.pdf